

MELFORT, SK, CanadaWMO: **714560**Lat: **52.8164N**Lon: **104.5964W**Elev: **1608**StdP: **13.86**Time Zone: **-6.00 (W06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-23.3	-33.7	0.9	-27.5	-28.8	1.2	-23.3	24.6	15.1	22.3	14.5	7.9	270	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	83.4	66.1	80.1	64.7	77.1	62.9	70.0	78.8	67.6	75.8	65.5	73.5	11.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	105.7	74.7	64.6	96.7	71.9	62.4	89.5	69.7	35.1	79.0	33.0	75.8	31.3	73.3	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.9	17.9	-33.9	89.9	4.7	3.6	-37.3	92.5	-40.0	94.5	-42.6	96.5	-46.0	99.1		
(5)			-34.1	74.8	4.7	3.0	-37.4	76.9	-40.2	78.6	-42.8	80.3	-46.2	82.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.8	2.4	5.9	17.9	35.9	50.0	59.5	63.6	61.9	53.0	38.2	20.6	7.2
	DBStd	24.79	15.84	13.97	13.96	11.19	9.15	6.71	5.40	6.36	8.54	9.13	12.55	14.62
	HDD50	6921	1476	1236	996	434	117	7	0	1	65	378	883	1328
	HDD65	11140	1941	1656	1461	872	470	187	91	136	370	830	1333	1793
	CDD50	1382	0	0	0	13	117	293	422	371	153	13	0	0
	CDD65	125	0	0	0	0	5	23	48	41	8	0	0	0
	CDH74	1425	0	0	0	5	104	264	443	464	142	3	0	0
	CDH80	307	0	0	0	1	23	54	76	121	32	0	0	0
Wind	WSAvg	9.1	9.2	9.2	9.4	9.8	10.2	9.2	8.1	8.1	9.1	9.3	9.1	9.0
Precipitation	PrecAvg	16.4	0.6	0.5	0.7	0.8	1.7	2.5	2.7	2.2	1.7	1.1	0.7	0.9
	PrecMax	26.1	1.3	1.4	1.5	2.7	6.0	7.0	5.8	5.9	3.3	3.7	1.7	1.6
	PrecMin	11.1	0.1	0.1	0.0	0.0	0.2	0.5	0.7	0.2	0.2	0.1	0.2	0.3
	PrecStd	3.4	0.3	0.3	0.4	0.6	1.5	1.6	1.4	1.3	0.9	0.8	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.7	37.1	48.1	70.6	83.2	86.2	86.0	89.1	84.7	70.5	51.1	39.3
		MCWB	35.0	34.6	41.2	51.3	57.7	65.9	71.9	66.2	62.9	55.6	44.1	35.9
	2%	DB	34.2	33.1	41.0	64.3	76.3	80.3	81.8	83.1	78.0	61.8	44.1	34.7
		MCWB	32.5	31.5	37.1	48.8	55.4	62.9	68.7	65.8	59.9	50.0	39.5	32.4
	5%	DB	30.0	29.7	38.2	58.4	71.7	76.6	79.0	79.1	73.0	56.4	39.8	30.3
		MCWB	28.5	28.5	35.3	46.6	53.6	61.0	66.8	64.7	58.4	47.6	36.7	28.6
	10%	DB	25.4	26.0	35.5	52.5	67.6	73.0	75.8	75.4	68.0	51.7	36.2	26.8
		MCWB	24.5	25.0	33.4	43.6	51.5	59.0	65.2	62.9	55.8	45.3	34.0	25.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.2	35.0	41.5	54.0	61.1	70.5	75.1	71.3	66.3	56.2	44.6	36.2
		MCDB	37.3	36.7	47.2	66.2	70.8	80.2	82.2	82.1	76.9	69.1	50.7	38.5
	2%	WB	32.5	31.6	37.6	50.8	57.8	66.4	71.4	68.4	62.5	51.7	40.1	32.5
		MCDB	34.2	33.1	40.5	61.2	71.8	75.7	79.4	78.0	73.9	58.9	43.6	34.6
	5%	WB	28.6	28.5	35.4	47.4	55.5	63.7	68.8	66.3	59.4	48.5	36.9	28.8
		MCDB	29.9	29.7	37.8	56.7	68.5	72.2	76.4	75.8	70.3	55.4	39.4	30.2
	10%	WB	24.4	25.0	33.5	44.3	53.1	61.3	66.6	64.4	56.7	45.8	34.0	25.8
		MCDB	25.4	26.0	35.4	52.1	65.0	69.2	73.5	73.1	66.4	50.9	36.1	26.9
Mean Daily Temperature Range	5% DB	MDBR	15.3	15.9	15.8	18.4	22.5	19.5	19.6	21.2	20.6	16.5	13.4	14.1
		MCDBR	16.9	15.5	15.1	27.3	28.5	25.4	24.1	26.9	28.5	24.8	17.2	16.5
		MCWBR	15.8	14.1	12.0	15.1	13.0	12.0	13.2	12.6	13.6	14.5	12.9	15.0
	5% WB	MCDBR	16.6	15.0	14.6	24.8	24.7	20.9	22.1	23.1	25.5	23.0	16.5	15.9
		MCWBR	15.7	13.7	11.8	15.0	12.5	12.0	13.9	12.7	13.3	14.1	12.8	14.6
Clear-Sky Solar Irradiance	taub		0.241	0.245	0.306	0.330	0.344	0.365	0.368	0.367	0.314	0.284	0.247	0.244
	taud		2.291	2.318	2.241	2.377	2.391	2.372	2.359	2.362	2.499	2.535	2.454	2.257
	Ebn at Noon		245	282	279	287	287	281	278	272	277	266	246	218
	Edh at Noon		19	25	34	34	35	36	36	34	26	20	16	16
All-Sky Solar Radiation	RadAvg		335	672	1171	1539	1842	1854	1914	1578	1118	642	306	237
	RadStd		32	38	86	129	156	141	114	124	108	102	39	30